A Lightweight Multiscale Fusion Framework for Traffic Vehicle Detection From Satellite Remote Sensing, UAV, and CCTV Imagery

Zhen Liu, Mei-Po Kwan, Weiwei Jiang, *Senior Member, IEEE*, Yang Liu, and Bingyan Cui

***Abstract*—Vehicle detection from heterogeneous traffic imagery is essential for large-scale traffic monitoring. However, satellite remote sensing, uncrewed aerial vehicle (UAV), and closed-circuit television (CCTV) images differ substantially in spatial resolution, viewing geometry, illumination conditions, and vehicle scale, making unified cross-source detection challenging, especially for small targets and unseen source-domain distributions. To address these issues, this study proposes a lightweight source-conditioned multiscale detection framework for vehicle detection from independent satellite, UAV, and CCTV image domains. The framework is centered on a source-aware multiscale awareness fusion module (SA-MS-AFM), which performs source-expert reweighting from a single input image and supports masked-expert training for incomplete expert availability. SwiftFormer with efficient additive attention (EAA) is adopted for efficient local–global representation, while Shiftwise convolution enhances local structural cues with low overhead. Content-aware reassembly of features (CARAFEs) is introduced in the neck to reduce spatial-detail loss during top-down feature reconstruction, and a four-scale prediction head is designed to accommodate large vehicle-scale variations. Experiments on a pooled cross-source dataset of 6805 images show that the proposed model achieves an F_1-score of 0.885, mAP@0.5 of 0.886, and mAP@0.5:0.95 of 0.513 while maintaining 41.53 frames per second (FPS) on the tested RTX 4090 GPU. Ablation, expert-masking, three-scale/four-scale tradeoff, and zero-shot external generalization tests further demonstrate the effectiveness and transferability of the proposed framework under heterogeneous traffic imaging conditions.**

***Index Terms*—Content-aware upsampling, cross-source image detection, multiscale feature fusion, source-aware gating, SwiftFormer, traffic vehicle detection.**

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I. Introduction

With the rapid development of cities and the continuous growth of vehicle ownership, traffic congestion and road safety problems have become increasingly prominent. Accurate and timely acquisition of traffic flow information is therefore essential. Vehicle detection is a key component of traffic information monitoring, which plays an important role in traffic flow estimation, congestion identification, violation monitoring, and emergency response [1]. Therefore, real-time and accurate vehicle detection has become an active research topic.

Traffic monitoring data are collected from heterogeneous sensing platforms, which differ substantially in spatial resolution, viewing geometry, illumination conditions, and vehicle scale [2]. Satellite remote sensing images have the advantage of wide coverage and stable acquisition and are suitable for large-area road network monitoring [3], [4]. However, their limited spatial resolution results in very small vehicle targets in the image, increasing the detection difficulty [5]. Uncrewed aerial vehicles (UAVs) imagery is flexible and suitable [6]. Therefore, it is suitable for road inspections, but it is also affected by flight altitude, weather conditions, buildings, and trees [7], [8]. Closed-circuit television (CCTV) images have the characteristics of high resolution and a large vehicle target size. It is widely distributed in urban traffic networks and is the main data source of traffic monitoring [9], but coverage blind spots still exist. These complementary characteristics indicate that a detector trained on a single image domain may not generalize well across different traffic sensing conditions [10]. Therefore, it is necessary to develop a unified cross-source vehicle detection framework that can adapt to multiple image domains, handle large-scale variations, and maintain efficient inference.

Recent advances in digital image processing (DIP) [11] and computer vision (CV) [12] have led to substantial advances in traffic vehicle detection. However, existing detection frameworks still face challenges when dealing with multiscale vehicle targets, small, weakly textured objects, and severe illumination or weather disturbances. Consequently, the development of models that combine multiscale adaptability, a

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lightweight framework, and effective cross-domain generalization is a crucial research issue.

II. Related Works

A. DIP-Based Traffic Vehicle Detection

Early research on traffic vehicle detection mainly focused on DIP applications to extract the moving foreground area from fixed surveillance video streams. Then, these foreground regions were classified using statistical learning methods. The mainstream approach relied on manual construction of complex target features.

In terms of background modeling and motion segmentation, the Gaussian mixture model (GMM) proposed by Stauffer and Grimson [13] laid the foundation for subsequent research. It uses a mixture of Gaussian distributions to model each pixel and allows the background representation to adapt to the dynamic changes in the scene. To cope with more complex traffic scenarios, the optical flow method was introduced into vehicle detection [14]. Barron et al. [15] demonstrated its effectiveness in capturing vehicle motion vectors, although its computational cost is relatively high. To improve detection speed and accuracy, hand-crafted feature-based classification frameworks became a mainstream method. Dalal and Triggs [16] introduced the histogram of oriented gradients feature combined with an SVM classifier. It improved detection performance by computing local histograms of gradient orientations. Felzenszwalb et al. [17] further proposed the deformable part model, which models a vehicle as a collection of parts and effectively addresses variations in viewpoint and partial occlusions. After that, Tang et al. [18] employed Haar features together with an AdaBoost cascade classifier for vehicle detection and achieved a significant speed improvement.

Although DIP-based methods have contributed to this field, they rely heavily on manually designed features and have difficulty distinguishing vehicles from complex backgrounds, such as tree occlusions and shadow interference. In addition, they often depend on sliding-window strategies, thereby introducing substantial redundant computation and limiting their applicability in network-level monitoring.

B. DL-Based Traffic Vehicle Detection

The introduction of convolutional neural networks (CNNs) has fundamentally changed the paradigm of traffic vehicle detection [19]. Early deep learning (DL)-based algorithms can be categorized into two-stage and one-stage methods.

Two-stage algorithms are represented by the region-CNN (R-CNN) family [20]. Faster R-CNN introduced the region proposal network (RPN). It enabled end-to-end training and achieved high detection accuracy in traffic monitoring scenarios [21]. To further improve the accuracy, various improvement methods, such as adopting a residual network (ResNet) backbone network [22] and feature pyramid networks (FPNs) [23], have been developed to better combine high-level semantic information with low-level geometric details. However, the computational cost of two-stage methods is still high and limits their network-level applications.

Due to the real-time requirements of traffic detection, single-stage algorithms have attracted more attention. The single-shot multibox detector (SSD) model [24], [25] has been used or modified to detect vehicles on feature maps and provides a practical balance between speed and accuracy [9]. The most representative one-stage framework, you only look once (YOLO), has also been widely applied to vehicle detection across its various versions [26], [27], [28]. For example, Keerthi Kiran et al. [29] used an improved YOLOv4 model with a multiscale Retinex method to detect traffic under different weather conditions. Ma et al. [30] improved the YOLOv5 model with Dense_CSPDarnet53 backbone and Focal-ECloU loss function to detect vehicles from UAV imagery. Sairam et al. [31] further developed a novel YOLOv8 with a non-maximum suppression strategy and improved the detection accuracy for traffic vehicles from UAV images. The inherent limitations of CNN architectures in capturing long-range dependencies have prompted researchers to explore the use of Transformers in vehicle detection. However, the standard vision Transformer (ViT) involves a high computational cost [32]. Tahir et al. [33] combined the YOLOv8 model with the Swin Transformer to improve the detection accuracy of small traffic vehicles in UAV imagery. AMSRDet further demonstrated the value of adaptive multiscale cross-modal fusion for UAV vehicle detection [34]. These studies achieved a substantial accuracy gain compared with the original YOLOv8 model and demonstrated the potential of integrating Transformer-based modules with one-stage detectors. Nevertheless, these studies remain restricted to UAV imagery and do not address cross-source vehicle detection across heterogeneous image domains.

Although advanced algorithms have made remarkable progress in specific scenarios [35], [36], [37], they still struggle to balance the lightweight design with efficient global feature awareness. Existing models are still limited in handling the domain discrepancy and scale variation among satellite remote sensing, UAV, and CCTV imagery. Most prior studies [38] focus on a single sensing platform or assume relatively consistent image distributions, whereas practical traffic monitoring often involves independent images collected from different platforms rather than co-registered observations of the same scene. In addition, existing models often lose critical spatial details during feature upsampling.

To address these issues, this study develops a lightweight source-conditioned multiscale framework for vehicle detection. It combines a SwiftFormer vision encoder with an enhanced YOLOv13 framework. We strengthen local representation learning with shiftwise convolution (SWConv) and exploit efficient additive attention (EAA) in SwiftFormer to improve local–global interactions. We further propose a source-aware multiscale awareness fusion module (SA-MS-AFM) with masked gating to perform source-expert reweighting and cross-resolution features, and adopt a content-aware reassembly of features (CARAFEs) upsampling structure to better preserve fine-grained spatial cues. In addition, a four-scale detection head is employed to accommodate the large variation in vehicle sizes. To improve practical robustness under incomplete source-expert availability, source

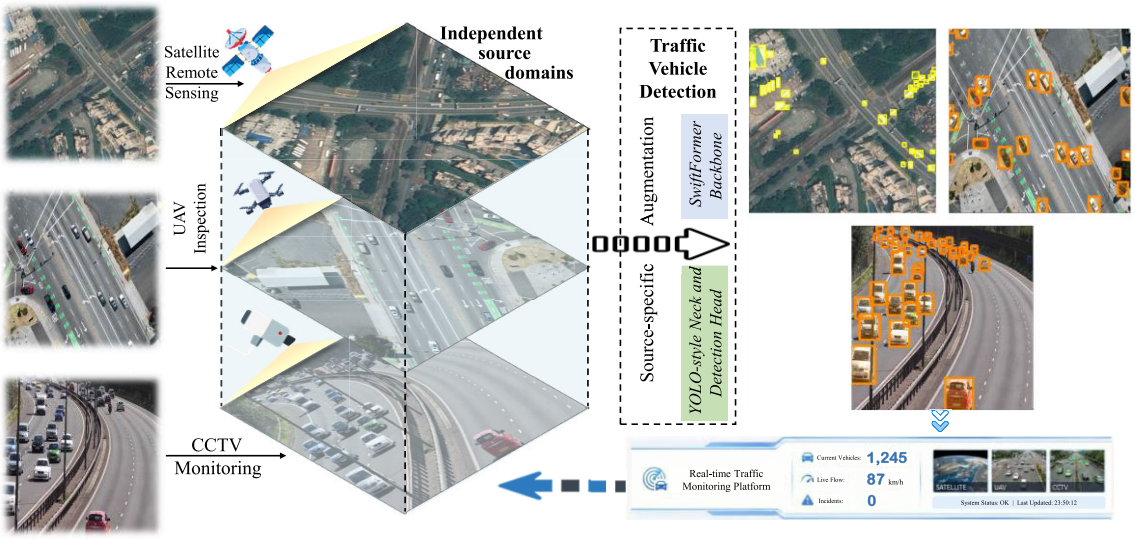


Fig. 1. Workflow of the lightweight source-conditioned multiscale fusion framework for traffic vehicle detection.

dropout is applied during training, together with a lightweight regularization on the source-aware weights. The main contributions of this study can be summarized as follows.

1) We establish a heterogeneous traffic vehicle dataset covering satellite remote sensing, low-altitude UAV inspection, and ground-based CCTV imagery for unified cross-domain evaluation. It provides a unified benchmark for training and evaluating cross-source vehicle detection models.
2) We propose a lightweight source-conditioned detection framework centered on SA-MS-AFM, which performs source-expert reweighting over independent satellite, UAV, and CCTV image domains. SWConv, CARAFE, and the four-scale head are integrated as supporting lightweight designs to improve local representation, detail recovery, and small-object detection.
3) Extensive quantitative experiments, source-expert masking analysis, ablation studies, and qualitative visualizations demonstrate the effectiveness, robustness, and real-time inference capability of the proposed framework.

III. Methodology

To address vehicle detection in heterogeneous traffic imagery, this study develops a lightweight source-conditioned multiscale fusion framework for satellite remote sensing, UAV, and CCTV images (see Fig. 1). Each input sample is a single image with a known source label, and no co-registered multisource observations are required. Source-specific data augmentation is first applied to enhance small-target visibility and reduce data imbalance. The augmented images are then processed by an optimized SwiftFormer backbone, where SWConv strengthens local structure modeling. SA-MS-AFM performs source-aware expert reweighting and cross-scale feature integration, while source dropout is applied during training to improve robustness when source-specific experts are unavailable. Finally, a lightweight YOLO-style neck with CARAFE upsampling and a four-scale detection head is used to preserve spatial details and handle large vehicle-scale variations with efficient inference.

A. SwiftFormer Backbone

SwiftFormer is a lightweight hybrid visual backbone that combines the efficiency of CNNs with global context modeling through an efficient attention design [39]. Its core idea is to alternately stack a Convolutional Encoder and an EAA module at each stage, enabling collaborative learning of local details and global dependencies with low computational complexity. As illustrated in Fig. 2, the network adopts a hierarchical four-stage design using increasing channel dimensions. This design supports multilevel representation learning from fine-grained textures to high-level semantics.

In the input stage, the image is embedded into an initial feature map F_0 through convolutional operations, which expands the feature dimension while reducing spatial resolution. Each SwiftFormer stage consists of a Convolutional Encoder and a SwiftFormer Encoder. The Convolutional Encoder captures local spatial patterns through depthwise separable and pointwise convolutions, helping preserve edge and texture cues at low computational cost. The SwiftFormer Encoder further enhances global context modeling through EAA. Unlike conventional self-attention, which requires the quadratic token-to-token attention matrix $\boldsymbol{Q}\boldsymbol{K}^T$, EAA generates a global query and performs elementwise query–key interaction, thereby reducing attention computation to a linear form, as formulated in (1). This design enables efficient long-range dependency modeling while maintaining lightweight inference

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{X}\boldsymbol{W}_q, \quad \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}_k \\ q_g &= \sum_{n=1}^{N} \text{Norm}\left(\frac{\boldsymbol{Q}w_g}{\sqrt{d}}\right)_g \boldsymbol{Q}_n \\ EAA(\boldsymbol{X}) &== W_o\left(q_g \odot \boldsymbol{K}\right) + \boldsymbol{Q} \end{aligned} \tag{1}$$

where $\boldsymbol{X} \in \mathbb{R}^{N\times C}$, $\boldsymbol{Q}$, $\boldsymbol{K} \in \mathbb{R}^{N\times d}$ are the projected query and key features, w_g is a learnable vector for generating

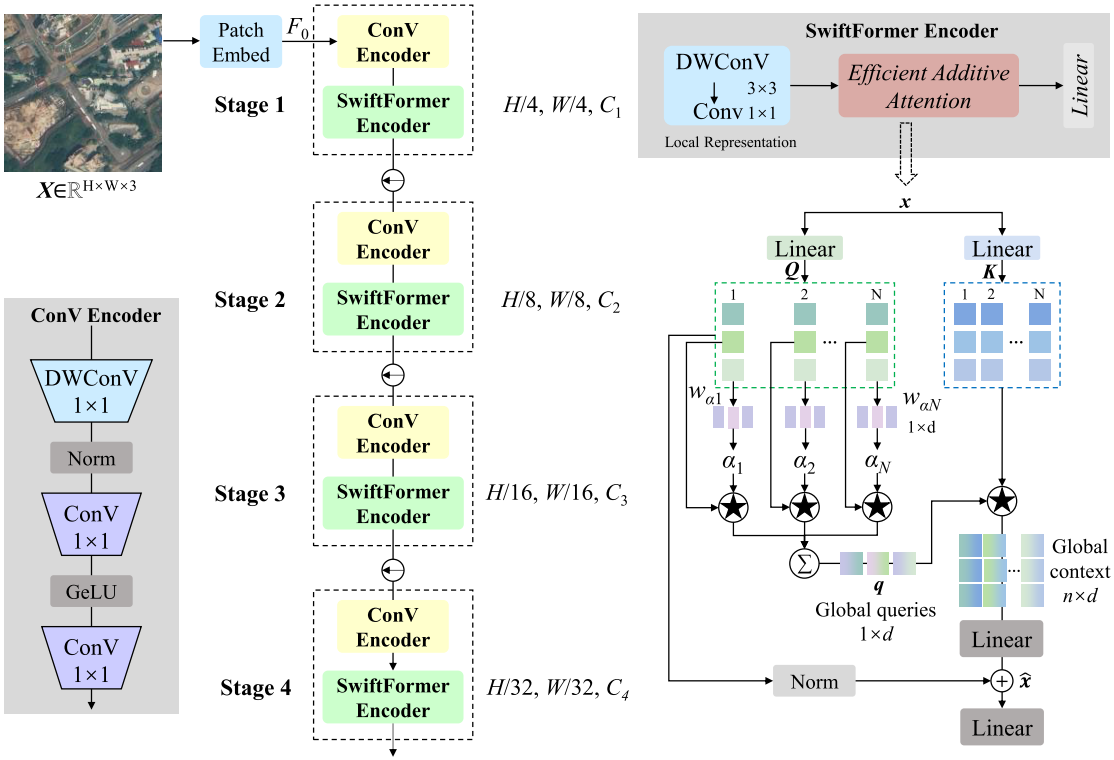


Fig. 2. Network architecture of the original SwiftFormer.

normalized global attention weights, q_g is the global query vector, $\odot$ denotes the elementwise multiplication, and W_o is the output projection. This formulation follows the SwiftFormer EAA design, which avoids the quadratic token-to-token attention matrix $\boldsymbol{QK}^T$ and achieves linear-complexity global interaction.

Between adjacent stages, convolutional downsampling is used to reduce spatial dimensions and increase channel capacity. The resulting multilevel features can be passed to downstream tasks through lightweight heads. In object detection, these backbones are typically coupled with multiscale neck structures to generate feature maps at different resolutions.

Although SwiftFormer performs well in general vision tasks, limitations arise in cross-source traffic vehicle detection. First, the extreme scale variation across satellite, UAV, and roadside imagery makes single-stage local–global fusion insufficient for small targets, particularly in high-resolution remote sensing images. Second, repeated downsampling may suppress fine spatial details that are critical for tiny vehicle instances. Third, multisource data differ substantially in viewpoint, illumination, and imaging quality, which can cause unstable cross-domain feature alignment when a single unified representation is used. Finally, practical traffic monitoring often involves incomplete sensing coverage, where certain sources may be unavailable due to operational constraints. These factors motivate a source-aware and multiscale fusion design that preserves spatial details and remains robust under incomplete source-expert availability.

B. Lightweight Source-Conditioned Fusion Framework for Cross-Source Traffic Vehicle Detection

We propose a lightweight source-conditioned feature fusion framework based on SwiftFormer to address vehicle detection across satellite remote sensing, UAV, and CCTV image domains with large variations in scale, viewpoint, and appearance. For each input sample, the model receives a single image I together with its source indicator $s\epsilon$ {*sat*, *uav*, *cctv*} or an availability mask m; no co-registered multisource observations are required. As shown in Fig. 3, the architecture employs a backbone–neck–head structure. In the backbone, we integrate the SWConv module to enhance local spatial modeling to capture weaker vehicle edges and structured backgrounds in high-resolution scenes. To explicitly deal with cross-source heterogeneity, we redesigned the SA-MS-AFM module as a source-expert reweighting module rather than a sensor-level fusion module. During training, source-expert dropouts randomly mask source-specific expert branches to improve robustness under incomplete source-expert availability. Source dropout does not remove an input modality or change the sampling distribution of source domains. Instead, it disables selected source-expert feature branches inside SA-MS-AFM while the input image remains unchanged. The module learns masked source-expert reliability weights and adaptively reweights feature branches conditioned on the input source. In the neck, we adopted a YOLO-style neck and introduced the CARAFE content-aware upsampling technique to reduce detail loss during top-down feature reconstruction. Finally, the detection head generates four-scale outputs aligned with the neck features, enabling scale-aware prediction from

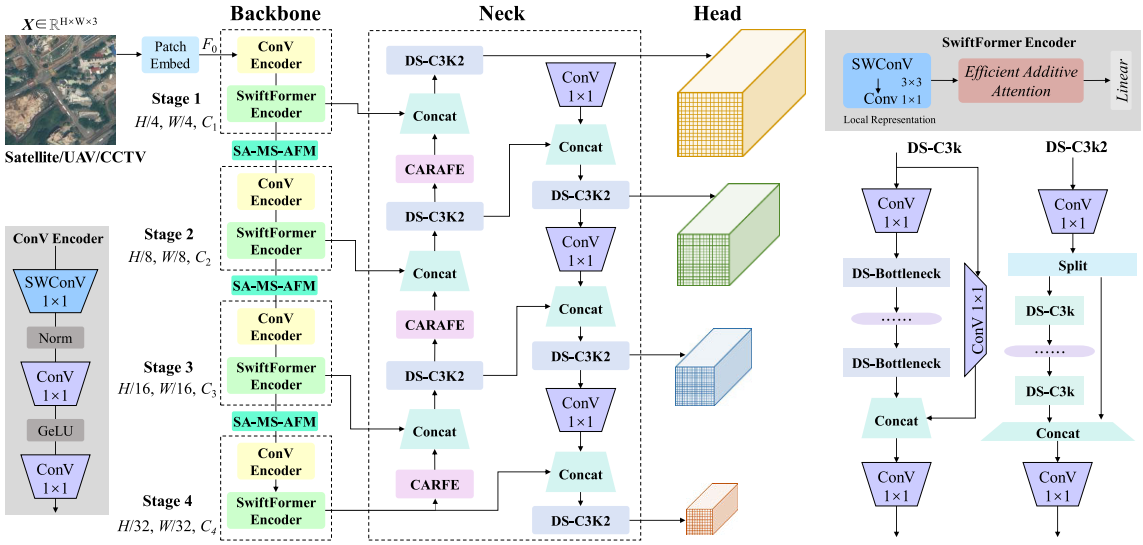


Fig. 3. Improved SwiftFormer network architecture for cross-source traffic vehicle detection.

tiny vehicles in satellite imagery to larger targets in UAV and CCTV imagery.

1) SWConv for Lightweight Local Context Enhancement: Satellite, UAV, and CCTV images exhibit substantial differences in viewpoint, illumination, texture, and vehicle scale. These differences often appear at local spatial scales and may propagate through the network, weakening subsequent source-conditioned feature reweighting. Since the framework targets high-resolution inputs, local contextual information should be enhanced with limited computational overhead. To this end, SWConv is introduced as a lightweight local representation enhancement module to replace the original depthwise convolution operations in the Convolutional Encoder and in the local representation branch of the SwiftFormer Encoder.

SWConv efficiently aggregates neighborhood information by reorganizing feature channels and combining shiftwise operations with grouped convolutions, thereby enabling cross-location information interaction. Compared with a standard 3×3 depthwise convolution, SWConv provides stronger local contextual representation while maintaining an efficient computation pattern. This helps preserve fine-grained texture and edge cues that are important for detecting small vehicles in high-resolution scenes.

As shown in Fig. 4, SWConv is embedded in two locations. First, in the Convolutional Encoder, it replaces the depthwise convolution that captures local spatial cues, thereby strengthening early stage representations before multiscale processing. Second, in the SwiftFormer Encoder, SWConv is used in the local representation branch to enhance local features that complement the global modeling performed by EAA. The attention pathway remains unchanged so that the efficient global interaction property of SwiftFormer is preserved.

By enhancing local context with low overhead, SWConv improves the discriminability of small vehicles. It also reduces sensitivity to cross-source appearance variations before source-conditioned expert reweighting. This design also supports high-resolution inference under real-time constraints.

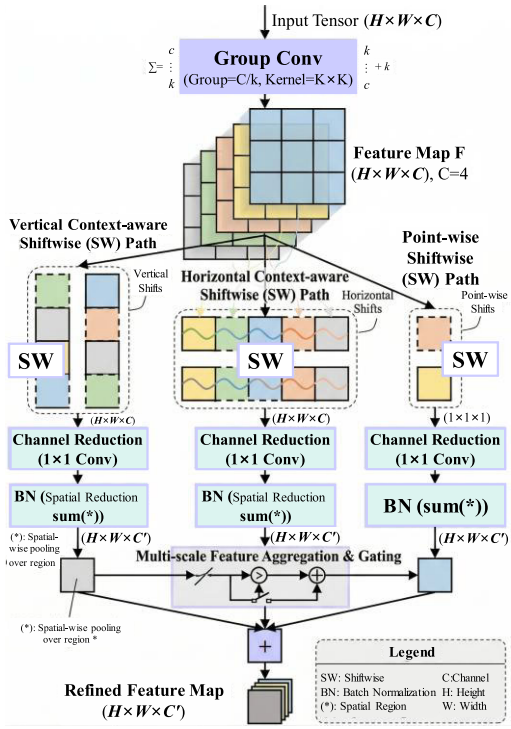


Fig. 4. Structure of the SWConv block used for efficient local context enrichment.

2) Source-Aware Multiscale Awareness Fusion Module: To address the heterogeneity across satellite remote sensing, UAV, and CCTV image domains, we extend the original multiscale awareness fusion design into a source-aware expert reweighting module, termed SA-MS-AFM, as shown in Fig. 5. In this study, each input sample is a single image with a known source label, and no co-registered satellite–UAV–CCTV observations are required. Therefore, SA-MS-AFM does not fuse synchronized sensor observations; instead, it learns the reliability of source-specific expert branches at each feature scale and reweights them adaptively. Source dropout is used during

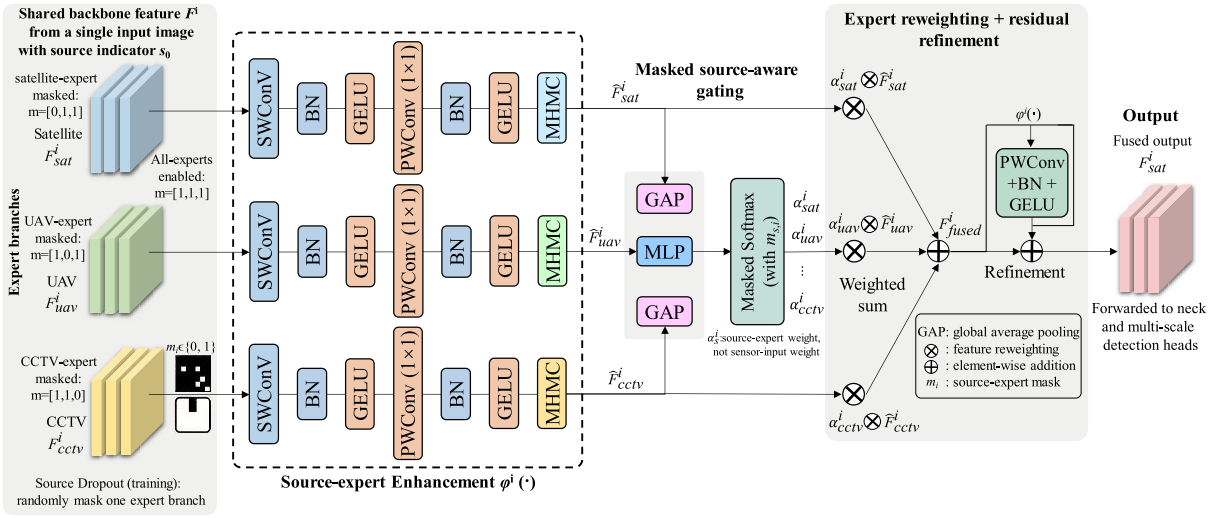


Fig. 5. Architecture of SA-MS-AFM for source-conditioned expert reweighting. The input is a single image feature map, and the masks m_s operate on source-expert branches rather than on synchronized sensor inputs.

training to randomly disable source-expert branches, thereby improving robustness when certain source domains or expert pathways are unavailable.

Let s ϵS ={*sat*, *uav*, *cctv*} be a source-expert branch, and let $i\epsilon$ {1, 2, 3, 4} be the feature scale. For an input image I from source domain s_0, the shared backbone produces a feature map F^i at scale i. SA-MS-AFM then applies parallel source-expert enhancement mappings to the same feature map F^i, producing branch-specific representations in the following equation:

$$\hat{F}^i_s = \phi^i_s\left(F^i\right), \quad s \in \{sat, uav, cctv\} \tag{2}$$

where F^i is the shared backbone feature of a single input image at scale i, and $\phi^i_s(\cdot)$ denotes the source-expert enhancement branch associated with source s. The three expert branches share the same architecture but are trained to capture source-related feature preferences. This source-expert transformation improves the discriminability of small vehicles in high-resolution satellite imagery and stabilizes local patterns under cross-source appearance variations.

Next, SA-MS-AFM performs masked source-aware gating over the source-expert branches. For each scale i, a compact gating function estimates a reliability logit z^i_s from the enhanced expert representation $\hat{F}^i_s$ and converts it into a normalized source-expert weight α^i_s, as shown in (3). Here, α^i_s denotes the relative contribution of the sth expert branch for a single input image, rather than the weight assigned to a synchronized sensor observation

$$\begin{aligned} z^i_s &= \mathrm{MLP}\left(\mathrm{GAP}\left(\hat{F}^i_s\right)\right) \\ \alpha^i_s &= \frac{m_s \exp\left(z^i_s\right)}{\sum_{s' \in S} m_{s'} \exp\left(z^i_{s'}\right) + \varepsilon} \end{aligned} \tag{3}$$

where GAP(·) denotes global average pooling, $m_s \epsilon$ {0, 1} is the mask indicating whether the sth source-expert branch is enabled, and ε is a small constant for numerical stability. During training, m_s is used for source dropout by randomly disabling one source-expert branch, which simulates incomplete source-domain availability or disabled expert pathways. The masked softmax normalization automatically renormalizes the weights over the enabled expert branches, improving the stability of source-conditioned feature reweighting during inference. To clarify that source dropout masks source-expert branches rather than removing input modalities or changing the source-domain sampling distribution, the detailed procedure of SA-MS-AFM is summarized in Fig. 6. Finally, the source-conditioned feature is obtained by weighted aggregation over the enabled source-expert branches and then refined

Algorithm 1 Source-Expert Dropout and Masked Reweighting in SA-MS-AFM

Require: Input image I; source label $s_0 \in \{\mathrm{sat}, \mathrm{uav}, \mathrm{cctv}\}$; backbone feature maps $\{F^i\}_{i=1}^4$; source-expert set $\mathcal{S} = \{\mathrm{sat}, \mathrm{uav}, \mathrm{cctv}\}$; dropout probability p_{sd}.

Ensure: Refined source-conditioned multi-scale features $\{\tilde{F}^i\}_{i=1}^4$.

Initialize expert mask $m_s \leftarrow 1$ for all $s \in \mathcal{S}$.
if training phase **then**
  Sample $u \sim \mathcal{U}(0, 1)$.
  if $u < p_{sd}$ **then**
    Randomly select one source-expert branch $r \in \mathcal{S}$.
    Disable the selected expert branch by setting $m_r \leftarrow 0$.
  end if
else
  Enable all expert branches for normal inference by setting $m_s \leftarrow 1$ for all $s \in \mathcal{S}$.
end if
for each feature scale $i \in \{1, 2, 3, 4\}$ **do**
  Obtain the shared backbone feature F^i of the single input image I.
  for each source-expert branch $s \in \mathcal{S}$ **do**
    Generate source-expert feature $\hat{F}^i_s \leftarrow \phi^i_s(F^i)$.
  end for
  for each source-expert branch $s \in \mathcal{S}$ **do**
    Compute reliability logit $z^i_s \leftarrow \mathrm{MLP}(\mathrm{GAP}(\hat{F}^i_s))$.
  end for
  Compute masked source-expert weights:

$$\alpha^i_s = \frac{m_s \exp(z^i_s)}{\sum_{s' \in \mathcal{S}} m_{s'} \exp(z^i_{s'}) + \varepsilon}$$

  Aggregate enabled source-expert features:

$$F^i_{\mathrm{agg}} = \sum_{s \in \mathcal{S}} \alpha^i_s \odot \hat{F}^i_s$$

  Apply lightweight residual refinement:

$$\tilde{F}^i = F^i_{\mathrm{agg}} + \varphi^i(F^i_{\mathrm{agg}})$$

end for
return $\{\tilde{F}^i\}_{i=1}^4$

Fig. 6. Training and inference procedure of source-expert dropout and masked reweighting in SA-MS-AFM.

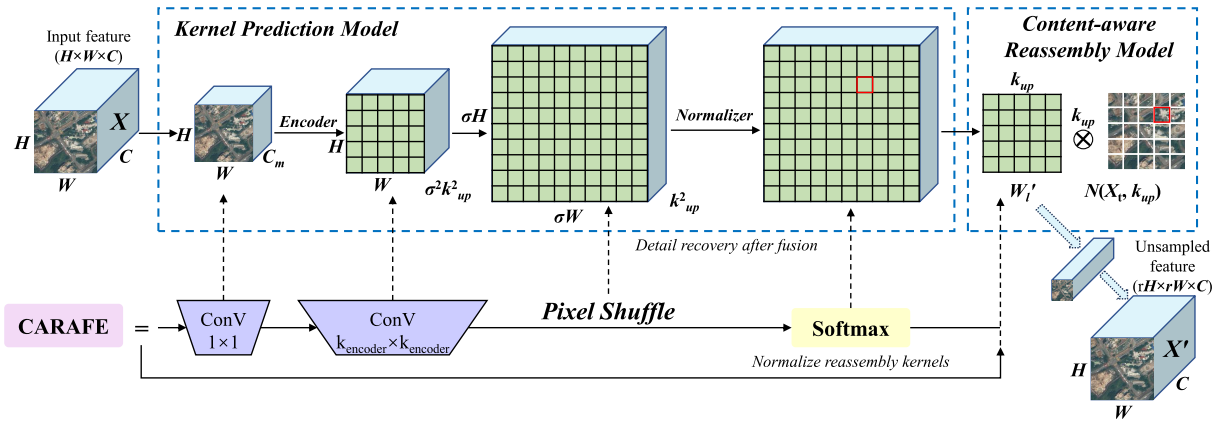


Fig. 7. CARAFE-based detail-preserving upsampling in the neck.

by a lightweight residual mapping, as given in the following equation:

$$F^i_{agg} = \sum_{s\in S} \alpha^i_s \hat{F}^i_s, \quad \tilde{F}^i = F^i_{agg} + \varphi^i\left(F^i_{agg}\right) \tag{4}$$

where F^i_{agg} is the aggregated expert feature at scale i, and φ^i is a lightweight refinement function implemented by pointwise convolution with normalization and activation. The refined feature $\tilde{F}^i$ is then forwarded to the neck and multiscale detection heads. It should be noted that the novelty of SA-MS-AFM does not lie in using GAP or MLP gating alone, but in its source-conditioned expert formulation for independent cross-source image domains, together with masked expert reweighting and source-expert dropout for incomplete expert availability.

3) CARAFE Module for Lightweight Upsampling: For cross-source vehicle detection, the neck aggregates multiscale features to support objects of different sizes. It is necessary to map high-level semantic features back to higher resolution formats to preserve detailed spatial information during top-down feature reconstruction. Conventional upsampling operators, such as nearest-neighbor or bilinear interpolation, are efficient but content-agnostic. They often smooth boundaries and dilute weak textures, which is unfavorable for tiny-vehicle detection. To address this issue, we adopt CARAFE as a lightweight upsampling module in the neck.

As described in Fig. 7, CARAFE performs upsampling through two stages: kernel prediction and content-aware reassembly. The kernel prediction module first compresses the channels of the input feature map $\boldsymbol{X}$ and then predicts a position-specific reassembly kernel. This process is expressed in the following equation:

$$\boldsymbol{W} = Soft\max(PixelShuffle(f_k(f_c(\boldsymbol{X})) \tag{5}$$

where $f_c(\cdot)$ denotes a 1×1 channel compression layer, $f_k(\cdot)$ denotes the kernel prediction encoder, and $\boldsymbol{W}$ is the normalized reassembly weight map. In the content-aware reassembly stage, the high-resolution feature map is reconstructed by aggregating local neighborhoods according to the predicted kernels in the following equation:

$$\boldsymbol{X}' = \text{Reassemble}(\boldsymbol{X}, \boldsymbol{W}) \tag{6}$$

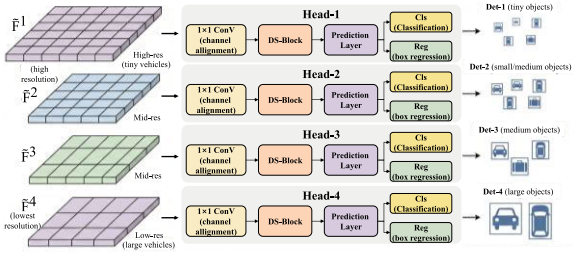


Fig. 8. Four parallel detection head branches for scale-aware vehicle prediction across heterogeneous traffic imagery.

where $\boldsymbol{X}'$ represents the reconstructed high-resolution feature map. Unlike fixed interpolation, CARAFE adapts the upsampling weights to local content, which helps retain edges and structures that are important for vehicle contours.

In our framework, CARAFE complements the source-conditioned detection pipeline by reducing spatial detail loss during top-down feature aggregation. This is critical because the reconstructed features need to support both tiny vehicles in satellite imagery and larger vehicles in UAV and CCTV imagery. Furthermore, CARAFE is lightweight and can be integrated into real-time pipelines. This property is particularly relevant for high-resolution inference settings in the high-resolution efficiency experiments.

4) Four-Scale Detection Head: To cope with the large-scale variation of vehicles across satellite, UAV, and CCTV imagery, the proposed framework adopts a four-scale detection head driven by the multilevel outputs of the neck, as illustrated in Fig. 8. The neck aggregates features from the four backbone stages through top-down and lateral connections. Specifically, deeper semantic features are upsampled using CARAFE and merged with shallower features through concatenation, followed by lightweight refinement blocks such as DS-C3K2 and 1×1 convolutions. This design produces four detection feature maps at different spatial resolutions, which are then fed into parallel prediction branches for multiscale detection.

Let $\tilde{F}^i (i = 1, 2, 3, 4)$ be the four source-conditioned neck outputs. Each $\tilde{F}^i$ is connected to an independent prediction branch $H_i(\cdot)$, forming a YOLO-style detection head [40]. Each branch generates classification and bounding-box regression outputs for objects best represented at that resolution. The

highest resolution is mainly beneficial for tiny vehicles, which frequently occur in satellite remote sensing imagery. The intermediate outputs support small- and medium-sized vehicles commonly observed in UAV imagery, while the lowest-resolution output helps detect larger vehicles that often appear in CCTV imagery. These assignments are scale-oriented rather than source-exclusive, since each branch can be activated for any input domain according to object size.

This four-scale design complements the source-conditioned detection pipeline in two ways. First, SA-MS-AFM provides source-conditioned representations before neck aggregation, helping reduce cross-source feature inconsistencies. Second, CARAFE-based upsampling enhances spatial detail recovery in the top-down pathway, which is particularly beneficial for the high-resolution detection layer targeting small objects. Overall, the four-scale detection head enables scale-aware prediction across heterogeneous traffic imagery and provides the basis for subsequent efficiency and ablation analyses.

5) Loss Function: The proposed network is optimized end-to-end with a multiscale detection objective defined over the four prediction layers. Let $\{Hi(\tilde{F}^i)\}_{i=1}^{4}$ be the outputs of the four-scale detection head, where each layer mainly responds to objects at a specific scale range. The overall detection loss is formulated as given in the following equation:

$$L_{\text{det}} = \sum_{i=1}^{4} \left(\lambda_{cls} L_{cls}^{(i)} + \lambda_{box} L_{box}^{(i)} + \lambda_{obj} L_{obj}^{(i)} \right) \tag{7}$$

where $L_{cls}^{(i)}$ is the classification loss, $L_{box}^{(i)}$ is the localization loss (IoU-based regression), and $L_{obj}^{(i)}$ supervises the objectness (foreground confidence) prediction. The coefficients λ_{cls}, λ_{box}, and λ_{obj} balance the contributions of the three terms. To improve robustness under incomplete source-domain availability, source dropout is applied during training by randomly disabling one source-expert branch in SA-MS-AFM. The masked source-aware gating renormalizes the expert weights over the enabled branches, and the detection loss in (7) is backpropagated through both the expert reweighting module and the detection head. This strategy encourages stable optimization when certain source-conditioned expert pathways are unavailable or suppressed during inference.

In addition, we introduce a lightweight regularization term on the source-expert weights α_i^s to stabilize training and prevent the gating distribution from becoming overly concentrated on a single expert branch. Specifically, an entropy regularization is adopted using the following equation:

$$L_{\text{gate}} = \frac{1}{4} \sum_{i=1}^{4} \sum_{s \in S} \alpha_s^i \log \left(\alpha_s^i + \varepsilon \right) \tag{8}$$

where $S = \{\text{sat, uav, cctv}\}$ is a small constant, and ε is a small constant for numerical stability. The final objective is given in the following equation:

$$L_{\text{total}} = L_{\text{det}} + \lambda_{gate} L_{gate}. \tag{9}$$

C. Data Augmentation Strategy

In this section, we applied different augmentation strategies based on the specific characteristics of the remote sensing

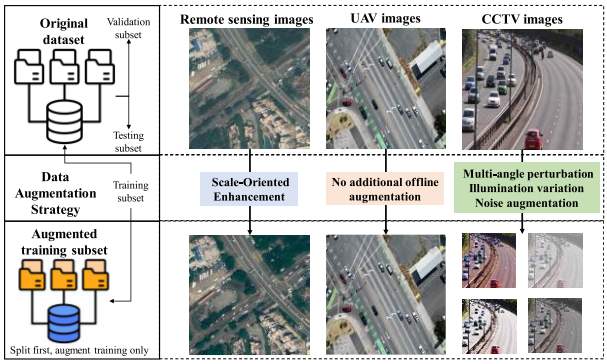


Fig. 9. Data split and augmentation protocol for the dataset.

(1636 images), UAV (2669 images from EVD4UAV dataset) [41], and CCTV (625 images from ROAD SEGMENTATION CV Model dataset) [42] datasets, as shown in Fig. 9. To avoid data leakage, all original images were first divided into training, validation, and testing subsets before any augmentation was performed. Augmentation was applied only to the training subset, while the validation and testing subsets contained only original images. For CCTV imagery, images from the same camera viewpoint, road segment, or highly similar scene were kept within the same subset where possible. For UAV imagery, frames from the same flight sequence or continuous road segment were not distributed across different subsets. For satellite imagery, geographically adjacent or overlapping patches were carefully separated to reduce spatial leakage between training and testing data.

1) Remote Sensing Scale-Oriented Training Augmentation: To address extremely small vehicle size and weak texture in remote sensing imagery, scale-oriented augmentation is applied to the training subset as follows.

1) *Scale Jittering:* Randomly enlarge the image to shorter dimensions (1.5–3.0 times) to enhance texture visibility.
2) *Local Magnification:* Crop and magnify local areas (2–3 times) to enhance edge clarity and structural information. This operation improves the prominence of features and prevents small objects from being obscured by background textures.

2) CCTV Training Augmentation and Enhancement: To address data scarcity and environmental variation, the following augmentation operations were applied only to the CCTV training subset after the original data split.

1) *Small-Angle Rotation ($\pm 10°$):* Enhance adaptability to different shooting angles and road structures.
2) *Brightness Adjustment (60%–120%):* Simulates complex lighting scenes such as glare or shadows.
3) *Gaussian Noise Injection (0.01–0.03):* Simulates sensor noise and compression artifacts [43].

Finally, the training, validation, and testing sets were constructed with an 8:1:1 split, as shown in Table I. The split was performed on original images before augmentation, and augmentation was then applied only to the training set. This protocol prevents augmented variants of the same original image from appearing simultaneously in training and evaluation subsets.

TABLE I
DATASET DIVISION OF MULTISOURCE TRAFFIC VEHICLE IMAGES

Image source	Training dataset (5443)		Validation dataset (681)		Testing dataset (681)		Total
	labeled	unlabeled	labeled	unlabeled	labeled	unlabeled	
Remote sensing	1029	279	123	41	127	37	1636
UAV	2018	117	248	19	249	18	2669
CCTV	2000	0	250	0	250	0	2500

(Note: The train/validation/test split was performed before augmentation. Augmentation was applied only to the training subset.)

TABLE II
PERFORMANCE AND COMPLEXITY COMPARISON ON THE POOLED CROSS-SOURCE VEHICLE TEST SET

Model			Performance Metrics					
Version	Params (M)	GFLOPs@1024	P	R	F_1-score	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv5s	7.2	42.2	0.835	0.812	0.823	0.828	0.462	52.37
YOLOv8s	11.2	73.7	0.851	0.826	0.838	0.842	0.476	48.19
YOLOv10s	7.3	53.3	0.847	0.837	0.842	0.841	0.484	49.86
YOLOv11s	9.4	55.1	0.849	0.833	0.841	0.838	0.472	47.69
YOLOv13s	9.0	53.1	0.863	0.842	0.852	0.854	0.492	46.73
YOLOv8s-MobileNetV3	5.2	26.7	0.791	0.766	0.778	0.762	0.388	73.68
Mamba vision-YOLO [46]	21.8	270.1	0.852	0.831	0.841	0.845	0.493	41.49
EfficientDet-D1	6.6	15.6	0.826	0.801	0.813	0.815	0.445	34.56
DETR-SPP [47]	41.2	145.0	0.842	0.817	0.829	0.832	0.462	27.72
EfficientFormer-L1	12.3	103.4	0.854	0.832	0.843	0.841	0.473	38.45
RT-DETR-MobileNetV4 [48]	11.3	101.1	0.858	0.831	0.844	0.846	0.477	37.86
MViT-S	26.1	211.2	0.857	0.835	0.846	0.845	0.479	29.69
Swin-Tiny	28.3	205.7	0.860	0.838	0.849	0.850	0.485	31.24
Ours	17.6	124.8	0.881	0.889	0.885	0.886	0.513	41.53

(Note: Params and GFLOPs were calculated or rescaled at 1024×1024. All baselines used public or reproduced standard implementations and followed the same training and evaluation protocols. YOLOv13s denotes a reproduced YOLOv13-style baseline without the proposed modules.)

IV. RESULTS AND DISCUSSION

A. Experimental Setup and Metrics

1) Experimental and Implementation Details: All tests were conducted on the same hardware platform to ensure a fair comparison. The experiments were performed on an NVIDIA RTX 4090 GPU with 24 GB of video memory, an Intel i9 CPU, and 64 GB of memory. The algorithm was implemented using the PyTorch DL-based framework, and CUDA 12.1 and cuDNN 8.6 were used.

To ensure consistent training and evaluation across heterogeneous source domains, all images were resized to 1024×1024 before being fed into the model. The model utilized the AdamW optimizer, starting with a learning rate of 1×10^{-4}, which was dynamically adjusted via a cosine schedule. The batch size was set to 8, and the number of training epochs was set to 150. In addition, early stopping was enabled during training to avoid unnecessary training and overfitting. Unless otherwise specified, all baseline models were trained and evaluated using the same leakage-controlled data split, in which original images were partitioned before augmentation, and only the training subset was augmented.

2) Model Configuration and Training Strategy: The proposed framework follows a backbone-source-expert reweighting neck–head pipeline. The backbone is built upon an enhanced SwiftFormer architecture with SWConv-based local representation, and the source-expert features are integrated by the proposed SA-MS-AFM. The neck adopts a YOLO-style multiscale aggregation design with CARAFE upsampling, and the detector outputs four-scale predictions to cover large vehicle-scale variation across satellite, UAV, and CCTV imagery. To ensure stable optimization of each component, the backbone is pretrained for ten epochs, followed by end-to-end training of the entire network. During training, source-expert dropout is applied by randomly masking one source-expert branch to simulate incomplete source-domain availability or disabled expert pathways. In addition, a lightweight regularization term is imposed on the source-expert gating weights to stabilize source-conditioned reweighting. Automatic mixed precision (AMP) training is employed to improve training efficiency, particularly under high-resolution inputs.

To ensure a fair comparison, all baseline detectors were trained and evaluated using the same leakage-controlled data split, input resolution, and training augmentation protocol. Unless otherwise specified, the models were initialized with publicly available pretrained weights and fine-tuned on the proposed dataset for 150 epochs using the AdamW optimizer, an initial learning rate of 1×10^{-4}, cosine learning-rate scheduling, and a batch size of 8. Frames per second (FPS) was measured only on the RTX 4090 platform using an input size of 1024×1024, batch size of 1, and averaged over the complete test set after warm-up iterations. Detailed baseline configurations are summarized in Table II.

3) Evaluation Metrics: Several evaluation metrics were adopted to assess detection accuracy and inference efficiency. Precision (P), Recall (R), and F_1-score were used to evaluate detection correctness, completeness, and their harmonic balance, respectively.

Average precision (AP) is calculated from the precision–recall curve [44]. Since we focus on single-class vehicle detection, mAP@0.5 is equivalent to AP@0.5 calculated over all vehicle instances at an IoU threshold of 0.5, as given in the following equation:

$$mAP@0.5 = AP_{0.5} = \int_0^1 P(R)dR, IoU = 0.5. \quad (10)$$

Then, mAP@0.5:0.95 is computed by averaging AP over IoU thresholds from 0.50 to 0.95 with a step of 0.05, as given in the following equation [45]:

$$mAP@0.5:0.95 = \frac{1}{10} \sum_{t\in\{0.50,0.55,\ldots,0.95\}} AP_t \quad (11)$$

where AP_t denotes the AP calculated at IoU threshold t. In addition, FPS is reported to measure inference efficiency under the specified input resolution and hardware settings.

B. Model Performance on Cross-Source Images

1) Test Results on the Pooled Cross-Source Test Set: We compared our model with representative baseline detectors on the pooled cross-source vehicle test set. As shown in Table II, the comparison covers YOLO-style one-stage detectors, lightweight CNN-based detectors, Mamba-enhanced detectors, DETR-style detectors, and Transformer-based detectors. Since the input size, training schedule, optimizer, augmentation strategy, and FPS protocol were kept identical for all models, Table II focuses on model-specific differences, including model version, Params, GFLOPs@1024, and detection performance.

Among the YOLO-series baselines, YOLOv13s achieves the best performance, with an F_1-score of 0.852, mAP@0.5 of 0.854, and mAP@0.5:0.95 of 0.492. Compared with YOLOv13s, the proposed model improves these metrics to 0.885, 0.886, and 0.513, respectively. Although the proposed model introduces additional parameters and computation due to SWConv, SA-MS-AFM, CARAFE, and the four-scale prediction head, it still maintains real-time inference at 41.53 FPS on the tested RTX 4090 GPU. This indicates that the proposed source-conditioned multiscale design improves detection accuracy without excessive efficiency loss.

Compared with lightweight real-time detectors, such as YOLOv8s-MobileNetV3 and EfficientDet-D1, the proposed model achieves substantially higher detection accuracy, although these lightweight models have smaller computational costs or higher FPS. For example, YOLOv8s-MobileNetV3 reaches 73.68 FPS but only obtains an mAP@0.5 of 0.762. In contrast, the proposed model improves mAP@0.5 to 0.886 while still satisfying the commonly used real-time threshold of 30 FPS.

Compared with Mamba-, DETR-, and Transformer-based detectors, including Mamba Vision-YOLO, DETR-SPP, RT-DETR-MobileNetV4, EfficientFormer-L1, MViT-S, and Swin-Tiny, the proposed model also achieves the best overall accuracy. Mamba Vision-YOLO and Swin-Tiny provide competitive detection performance, but they generally involve higher computational cost and/or lower inference speed. Overall, the proposed framework achieves the highest F_1-score, mAP@0.5, and mAP@0.5:0.95 among all compared detectors on the pooled cross-source test set while maintaining a favorable balance between accuracy, complexity, and GPU-level inference efficiency.

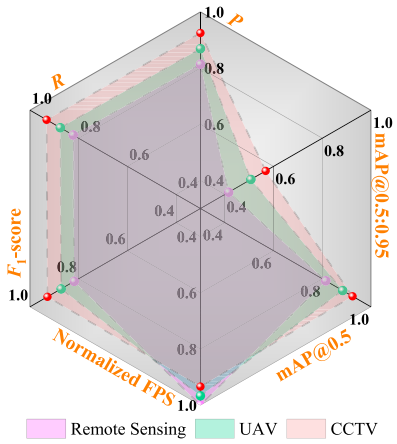


Fig. 10. Detection performance radar chart for single-source images.

2) Test Results Across Individual Source Domains: To further analyze our model's performance on each source domain, Fig. 10 provides a radar-chart visualization for six metrics (P, R, F_1-score, mAP@0.5, mAP@0.5:0.95, and normalized FPS). The three polygons reveal their respective unique and consistent performance patterns, which agree well with the imaging features of each data source.

For remote sensing images, the model achieved an F_1-score of 0.818 and an mAP@0.5 of 0.815, indicating that this source remains the most challenging because vehicles occupy only a few pixels and are easily affected by complex backgrounds. Nevertheless, the normalized FPS reaches 1.000, showing that the proposed framework maintains the highest inference efficiency on this subset while still preserving acceptable detection accuracy for tiny targets.

For UAV images, all detection metrics are significantly improved, with an F_1-score of 0.872. This improvement can be attributed to the more favorable vehicle scale and richer texture information, which allow the source-aware fusion and CARAFE upsampling modules to exploit cross-scale context more effectively. Meanwhile, the normalized FPS is 0.967, indicating that the model preserves high inference efficiency while achieving a balanced and robust performance on UAV imagery.

For CCTV images, the model achieved the best detection accuracy, with an F_1-score of 0.928 and the highest mAP-related metrics, because the vehicles appear larger and are visually more distinguishable. The normalized FPS is 0.934, which is slightly lower than that of the other two sources but still indicates real-time performance. This small reduction is acceptable and may be related to denser traffic scenes and more active predictions during detection.

C. Source-Domain Robustness, Expert Masking, and Efficiency Analyses

To avoid ambiguity between source-domain evaluation and masked-expert robustness, we separate the analysis into two

TABLE III
DOMAIN-WISE DETECTION PERFORMANCE ON THE POOLED AND INDIVIDUAL SOURCE-DOMAIN TEST SUBSETS

Model	Pooled test set		Satellite-domain		UAV-domain		CCTV-domain	
	F_1-score	mAP@0.5	F_1-score	mAP @0.5	F_1-score	mAP @0.5	F_1-score	mAP @0.5
YOLOv13s	0.852	0.854	0.785	0.791	0.837	0.834	0.872	0.885
Mamba vision-YOLO	0.841	0.845	0.772	0.784	0.828	0.833	0.875	0.879
Swin-Tiny	0.849	0.850	0.791	0.765	0.834	0.829	0.881	0.864
Ours	0.885	0.886	0.818	0.815	0.872	0.884	0.928	0.925
Ours w/o source dropout	0.879	0.882	0.801	0.776	0.852	0.861	0.895	0.887

TABLE IV
LEAVE-ONE-EXPERT-OUT MASKING ANALYSIS OF SA-MS-AFM

Setting	Expert mask	F_1-score	mAP@0.5	ΔF_1-score	ΔmAP@0.5
All experts enabled	[1,1,1]	0.885	0.886	-	-
Satellite-expert masked	[0,1,1]	0.882	0.884	-0.003	-0.002
UAV-expert masked	[1,0,1]	0.879	0.881	-0.006	-0.005
CCTV-expert masked	[1,1,0]	0.875	0.878	-0.010	-0.008
Random expert masked	random one zero	0.879	0.881	-0.006	-0.005

protocols. The first protocol is domain-wise evaluation, where the model is tested on the pooled cross-source test set and on individual source-domain subsets, including satellite remote sensing, UAV, and CCTV imagery. In this protocol, "satellite-domain," "UAV-domain," and "CCTV-domain" denote test subsets from different image domains, rather than missing-modality settings. The second protocol is source-expert masking, where the input image remains unchanged, but one source-expert branch in SA-MS-AFM is disabled during inference. Therefore, the missing-source setting in this study refers to incomplete source-expert availability inside the model, rather than missing co-registered multisensor observations.

1) Domain-Wise Evaluation Across Source Domains: Table III reports the F_1-score and mAP@0.5 results for each model under these settings. Overall, the compared models show consistent difficulty differences across individual source domains. The satellite-domain subset is the most challenging, followed by the UAV-domain subset, while the CCTV-domain subset is the easiest. Taking YOLOv13 as an example, its F_1-score and mAP@0.5 are 0.785 and 0.791 on the satellite-domain subset, while they increase to 0.872 and 0.885 on the CCTV-domain subset. This is mainly caused by the smaller vehicle footprint, weaker texture, and more complex background in satellite imagery.

On the pooled test set, the proposed method achieves the best overall performance (F_1-score = 0.885, mAP@0.5 = 0.886), outperforming YOLOv13s (0.852/0.854), Mamba-YOLO (0.841/0.845), and Swin-Transformer (0.849/0.850). When evaluated on individual source domains, the proposed method also remains superior, reaching 0.818/0.815, 0.872/0.884, and 0.928/0.925 for satellite-domain, UAV-domain, and CCTV-domain settings, respectively. These results indicate that the source-conditioned multiscale design improves adaptability across heterogeneous source-domain test subsets.

2) Source-Expert Masking Analysis Under Incomplete Expert Availability: To further evaluate the robustness of SA-MS-AFM under incomplete expert availability, we conducted a leave-one-expert-out masking analysis. Different from the domain-wise evaluation in Table III, this experiment keeps each test image unchanged and disables one source-expert branch in SA-MS-AFM during inference by setting the corresponding mask value to zero. The remaining branches are renormalized through masked softmax. The expert mask is ordered as [m_{sat},m_{uav}, m_{cctv}]. Therefore, this experiment evaluates masked source-expert reweighting rather than missing synchronized sensor inputs.

As shown in Table IV, the all-expert setting [1,1,1] achieves an F_1-score of 0.885 and an mAP@0.5 of 0.886. Masking the satellite-, UAV-, and CCTV-expert branches reduces the performance to 0.882/0.884, 0.879/0.881, and 0.875/0.878, respectively. The largest degradation occurs when the CCTV-expert branch is masked, which is consistent with its higher weights and indicates that high-resolution local cues contribute strongly to detection. However, the model still maintains stable performance under CCTV-expert masking, suggesting that it does not solely depend on the CCTV expert and that the remaining expert branches can provide complementary representations.

The randomly masked expert setting obtains an F_1-score of 0.879 and an mAP@0.5 of 0.881, showing a moderate decrease from the all-expert setting. Overall, these results demonstrate that source-expert dropout improves the stability of masked-expert inference and reduces the risk of excessive dependency on a single expert branch.

3) High-Resolution Efficiency Analysis: To evaluate inference efficiency under high-resolution traffic monitoring scenarios, we further measured the FPS of each comparative model at four input resolutions: 640, 768, 1024, and 1280 pixels. As shown in Fig. 11, the FPS of all models decreases as the

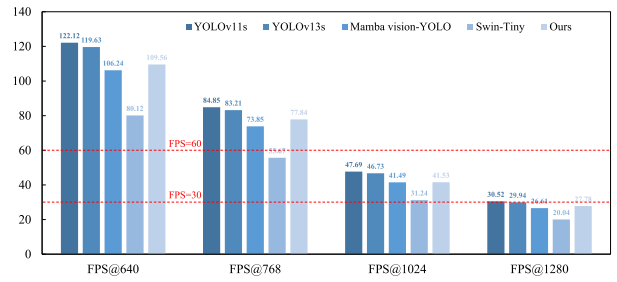


Fig. 11. FPS comparison of different models under different input resolutions.

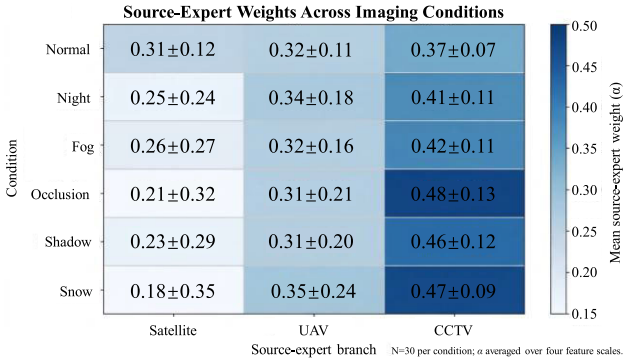


Fig. 12. Heatmap of source-expert weights across imaging conditions.

input resolution increases, which is consistent with the higher computational cost introduced by larger feature maps.

At input sizes of 640 and 768 resolutions, most detectors maintain high inference speeds, indicating their suitability for real-time monitoring under moderate-resolution inputs. At 1024 resolution, the speed differences become more representative. YOLOv11 and YOLOv13s still achieve approximately 47 FPS, while our model reaches 41.53 FPS, which is higher than Swin-Tiny (31.24 FPS) and close to Mamba-YOLO (41.49 FPS). This shows that the proposed framework preserves competitive efficiency while improving detection accuracy. When the input resolution increases to 1280, the FPS of all models drops substantially. Some models fall below the commonly used 30 FPS real-time threshold, while the proposed model maintains 27.78 FPS, indicating near-real-time inference capability under very high-resolution inputs.

Overall, Fig. 11 demonstrates that the framework achieves a favorable balance between detection accuracy and GPU-level inference efficiency across different input resolutions. These results support its potential for high-resolution traffic monitoring tasks on GPU-supported computing platforms.

It should be noted that the reported FPS results were measured on an RTX 4090 GPU and should not be directly interpreted as edge-device performance. Deployment on constrained platforms such as Jetson devices or mobile GPUs may require model compression, TensorRT/FP16/INT8 acceleration, and hardware-specific optimization, which will be investigated in future work.

4) Source-Expert Weight Interpretation and Dependency Bias Analysis: To further understand the source-conditioned expert reweighting mechanism of SA-MS-AFM, we analyze the distribution of source-expert weights under different imaging conditions, as shown in Fig. 12. For each condition, $N = 30$ samples were used, and the reported weights are averaged over four feature scales. Overall, the CCTV-expert branch generally receives higher weights under all conditions (0.37–0.48), suggesting that this branch provides more discriminative local cues when the vehicle targets are clear and relatively large. The UAV-expert branch remains at an intermediate level (0.31–0.35), indicating its complementary role in capturing road-segment-level vehicle profiles and local structures. The satellite-expert branch has relatively lower weights (0.18–0.31), especially at night, under occlusion, shadow, and snow conditions, reflecting the difficulty of extracting reliable cues from tiny vehicles with weak textures and complex backgrounds.

Meanwhile, the standard deviation of weights increases under challenging conditions, such as night, fog, occlusion, and snow, indicating that the model tends to adjust expert-branch dependency according to sample quality and visibility. This adaptive behavior helps stabilize source-conditioned feature reweighting under degraded imaging conditions.

Although the CCTV-expert branch receives the highest weights, the leave-one-expert-out results in Table IV show that the model does not simply rely on this branch. Masking the CCTV-expert branch causes the largest performance drop, from 0.885/0.886 to 0.875/0.878 in F_1-score/mAP@0.5, confirming its strong contribution to high-resolution local cues. However, the performance remains stable after masking, and the randomly masked expert setting still achieves 0.879/0.881. These results indicate that the satellite-, UAV-, and CCTV-expert branches provide complementary representations rather than forming an excessive dependency on a single expert.

D. Ablation Experiments and Component Analysis

Ablation experiments were conducted to evaluate the contribution of each component. As shown in Table V, the baseline model was progressively enhanced by introducing the SwiftFormer backbone, SWConv, SA-MS-AFM, CARAFE upsampling, four-scale detection head, source-expert dropout, and gate regularization. To better disentangle the contribution of the final components, the four-scale head, source-expert dropout, and gate regularization were added separately in variants E1, E2, and E3/Ours.

The baseline model achieved an F_1-score of 0.852, mAP@0.5 of 0.854, and mAP@0.5:0.95 of 0.492 on the pooled cross-source test set, with an inference speed of 46.7 FPS. After replacing the backbone with SwiftFormer (A), the performance improved to 0.861, 0.862, and 0.496, indicating that the lightweight hybrid encoder improves feature representation under heterogeneous source domains. After further adding SWConv (B), the metrics increased to 0.864, 0.869, and 0.500, showing that SWConv strengthens local context modeling with limited computational overhead. The introduction of SA-MS-AFM (C) brought more obvious gains, improving the F_1-score and mAP@0.5 to 0.874 and 0.877, respectively. This demonstrates the effectiveness of source-aware expert reweighting and multiscale feature integration in reducing cross-source scale discrepancy and representation inconsistency. Subsequently, adding

TABLE V
ABLATION RESULTS OF THE PROPOSED SOURCE-CONDITIONED DETECTION FRAMEWORK

Model variant	SwiftFormer backbone	SW Conv	SA-MS-AFM	CARAFE	Four-Scale head	Source-expert dropout	Gate reg.	F_1-score	mAP @0.5	mAP@ 0.5:0.95	FPS
Baseline	✗	✗	✗	✗	✗	✗	✗	0.852	0.854	0.492	46.70
A	✓	✗	✗	✗	✗	✗	✗	0.861	0.862	0.496	45.90
B	✓	✓	✗	✗	✗	✗	✗	0.864	0.869	0.500	45.20
C	✓	✓	✓	✗	✗	✗	✗	0.874	0.877	0.506	44.00
D	✓	✓	✓	✓	✗	✗	✗	0.880	0.882	0.509	43.00
E1	✓	✓	✓	✓	✓	✗	✗	0.882	0.884	0.511	41.56
E2	✓	✓	✓	✓	✓	✓	✗	0.884	0.885	0.512	41.54
E3/Ours	✓	✓	✓	✓	✓	✓	✓	0.885	0.886	0.513	41.53
E'	✓	✓	✓	✓	✓	✗	✓	0.879	0.882	0.508	41.53

TABLE VI
THREE-SCALE AND FOUR-SCALE DETECTION HEAD TRADEOFF ANALYSIS

Head configuration	F_1-score	mAP@0.5	mAP@0.5:0.95	AP_s	FPS	Params (M)	GFLOPs@1024	GPU memory (GB)
Three-scale head	0.881	0.883	0.509	0.468	43.12	16.9	116.4	6.21
Four-scale head	0.885	0.886	0.513	0.521	41.53	17.6	124.8	6.74

(Note: AP_s denotes the AP for the smallest one-third of vehicle instances ranked by bounding-box area.)

CARAFE (D) further improved the model to 0.880, 0.882, and 0.509, indicating that content-aware upsampling alleviates spatial detail loss during top-down feature reconstruction.

To isolate the contribution of the final components, the four-scale detection head was first added to variant D, forming E1. Compared with D, E1 improved the F_1-score, mAP@0.5, and mAP@0.5:0.95 to 0.882, 0.884, and 0.511, respectively, while the FPS decreased from 43.00 to 41.56 due to the additional high-resolution prediction branch. Then, source-expert dropout was introduced in E2, further improving the metrics to 0.884, 0.885, and 0.512, with nearly unchanged inference speed. Finally, gate regularization was added in E3/Ours, achieving the best overall performance, with an F_1-score of 0.885, an mAP@0.5 of 0.886, and an mAP@0.5:0.95 of 0.513 while maintaining 41.53 FPS on the tested RTX 4090 GPU. These results show that the four-scale head, source-expert dropout, and gate regularization each provide incremental but distinguishable contributions.

In addition, Table V includes the control variant E', in which source-expert dropout is removed while the other structures remain unchanged. The performance of E' decreased to an F_1-score of 0.879, mAP@0.5 of 0.882 while maintaining the same inference speed. This indicates that source-expert dropout improves the stability of masked-expert inference and reduces performance degradation when one expert branch is disabled, which is consistent with the results in Table IV.

Since the four-scale detection head introduces an additional high-resolution prediction branch, we further compare the three-scale and four-scale configurations in Table VI while keeping the other modules unchanged. Compared with the three-scale head, the four-scale head improves the F_1-score, mAP@0.5, and mAP@0.5:0.95 from 0.881, 0.883, and 0.509 to 0.885, 0.886, and 0.513, respectively. More importantly, AP_s increases from 0.468 to 0.521, indicating that the additional high-resolution branch mainly benefits small-vehicle detection.

TABLE VII
EXTERNAL CROSS-DATASET GENERALIZATION RESULTS

Dataset	Source domain	F_1-score	mAP @0.5	mAP @0.5:0.95
VehSat	Satellite	0.805	0.801	0.395
UAV Images of Road Vehicles	UAV	0.856	0.849	0.488
TrafficCAM	CCTV	0.906	0.911	0.543

Although the four-scale configuration slightly increases the parameters from 16.9M to 17.6M, GFLOPs@1024 from 116.4 to 124.8, and GPU memory usage from 6.21 to 6.74 GB, the inference speed remains 41.53 FPS. Therefore, the fourth prediction branch represents a justified accuracy–efficiency tradeoff rather than an accuracy-only extension.

E. Zero-Shot External Generalization Test

To further evaluate the transferability of the proposed framework to unseen data distributions, we conducted a zero-shot external generalization test on three public datasets corresponding to the three source domains: VehSat for satellite imagery, UAV images of road vehicles for UAV imagery, and TrafficCAM for CCTV-style traffic imagery. The model was trained only on the proposed pooled cross-source dataset and was directly evaluated on these external datasets without fine-tuning. Vehicle-related categories in each dataset were merged into a single vehicle class to match the single-class vehicle detection setting of this study.

As shown in Table VII, the proposed model maintains stable performance across all external datasets, achieving F_1-scores of 0.805, 0.856, and 0.906 on VehSat [49], UAV images of road vehicles [50], and TrafficCAM [51], respectively. The results show a consistent difficulty order, with satellite

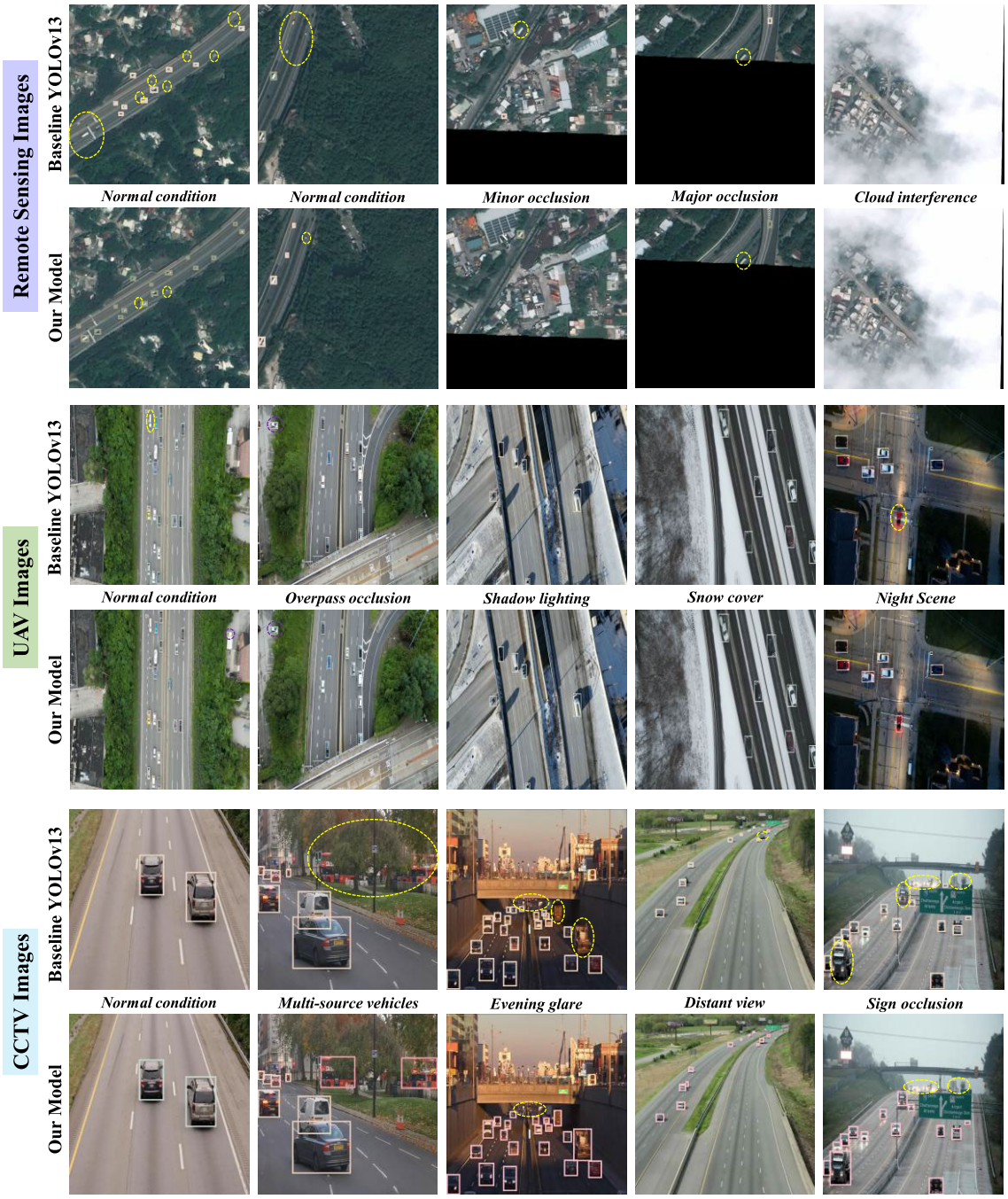


Fig. 13. Qualitative detection results across heterogeneous source domains and imaging conditions.

imagery being the most challenging, followed by UAV and CCTV imagery. Although external performance is lower than the internal test results due to differences in sensor platform, viewpoint, geographic region, and annotation protocol, these results indicate that the proposed source-conditioned multiscale design retains transferability under unseen source-domain distributions.

F. Visualization and Practical Implications

1) Qualitative Visualization Across Conditions: Fig. 13 presents a qualitative comparison between the YOLOv13 baseline model and the proposed model across different source domains and imaging conditions. Rectangular bounding boxes indicate detected vehicles, yellow ellipses denote missed detections, and purple ellipses represent false positives.

For remote sensing images, both models face challenges when processing small vehicle objects. Under normal conditions, YOLOv13 produces more missed detections, while our model detects more vehicles. This disparity becomes more pronounced when local occlusions occur. The improved model shows a more stable detection response and fewer missed detections. However, this advantage becomes less obvious in cloud-affected scenarios. Severe disturbances caused by atmospheric effects can limit the available texture information, which can affect the performance of both models.

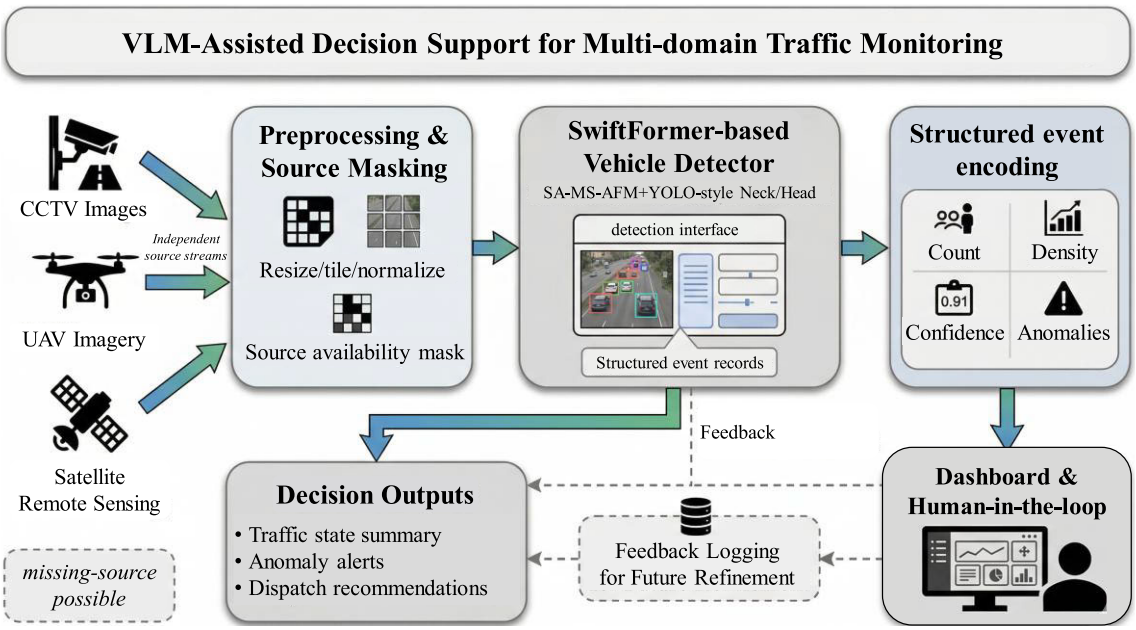


Fig. 14. Conceptual future direction of VLM-assisted decision support for multidomain traffic monitoring.

For UAV images, the overall performance is improved due to higher object resolution and better viewing angle. Both models correctly detect most vehicles under normal daytime conditions. However, in more complex scenes, such as night scenes, YOLOv13 has a higher missed detection rate. Our model maintains better consistency and successfully detects vehicles that the baseline model fails to detect. This improvement shows that the source-conditioned multiscale representation is more robust under low-light and partially occluded conditions.

For CCTV images, vehicles usually appear larger and have clearer contours; therefore, both models perform well under normal conditions. However, under unfavorable conditions, the performance gap becomes more apparent. Under conditions of varying illumination or partially obscured road signs, YOLOv13 generates more false positives and occasionally misses detections. The proposed model shows greater stability and maintains more accurate detection across diverse CCTV scenes.

2) Future Perspective: VLM-Assisted Decision Support for Traffic Monitoring: To further improve the interpretability and decision support capability of heterogeneous traffic monitoring systems in future applications, we propose a conceptual vision-language model (VLM) assisted decision fusion framework for future applications, as shown in Fig. 14. The framework decouples the perception layer from the decision layer without changing the main structure of our model.

1) First, images collected from satellite remote sensing, UAV, or CCTV platforms could be independently processed through preprocessing and source-mask generation, including resolution normalization, tiling or frame extraction, and source-availability mask construction.
2) Subsequently, each preprocessed image could be fed into the proposed source-conditioned vehicle detector, including SA-MS-AFM and the YOLO-style neck/head. The detector may provide bounding boxes, confidence scores, multiscale prediction results, and structured event records such as vehicle count, density estimation, confidence statistics, and anomaly candidates.
3) Next, VLM/LLM could combine rule constraints, prompt templates, structured event records, and contextual traffic information to support decision-level interpretation. It may generate traffic status summaries, abnormal-event alarms, or dispatch suggestions by integrating information from different source domains and temporal observations.
4) Finally, human feedback could be recorded to support future prompt refinement, rule optimization, or lightweight model updating.

This conceptual perspective suggests a possible decision-support pathway for extending the proposed detector to road-segment, road-network, and regional traffic monitoring applications. However, the implementation, quantitative evaluation, and human-in-the-loop validation of this VLM-assisted component are beyond the scope of the present study and will be investigated in future work.

V. Conclusion

In this study, a source-conditioned multiscale traffic vehicle detection framework was proposed for heterogeneous traffic monitoring imagery, including satellite remote sensing, UAV, and CCTV images. The framework is centered on SA-MS-AFM, which performs source-aware expert reweighting and multiscale feature integration for independent cross-source image domains. SwiftFormer, SWConv, CARAFE, and the four-scale prediction head are integrated as supporting designs to enhance local–global representation, preserve spatial details, and accommodate large vehicle-scale variations across different source domains.

The experimental results on a pooled cross-source dataset of 6805 images show that our model achieved an F_1-score of 0.885, mAP@0.5 of 0.886, and mAP@0.5:0.95 of 0.513, outperforming the YOLOv13 baseline while maintaining a real-time inference speed of 41.53 FPS on the tested RTX 4090 GPU. Compared with representative YOLO variants, lightweight CNN detectors, and Transformer-based baselines, the proposed method achieves a favorable balance between detection accuracy, model complexity, and GPU-level inference efficiency. The zero-shot external generalization test on three public datasets further indicates that the model retains transferability under unseen satellite, UAV, and CCTV data distributions. Ablation experiments, source-expert masking analysis, and three-scale/four-scale tradeoff analysis further verify the contribution of SA-MS-AFM, source-expert dropout, CARAFE, and the four-scale prediction head. Qualitative visualization results also demonstrated that the improved framework provides more stable detection under challenging conditions such as tiny targets, weak illumination, and occlusion.

In future work, we will further investigate vehicle discrimination in complex nonroad scenarios and explore the integration of the proposed detector with high-level semantic guidance, VLMs, and prompt engineering to improve scene understanding and application-level generalization. We will also evaluate and optimize the proposed framework on resource-constrained edge platforms to further verify its practical deployment capability.

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